IOT-ENABLED RFID FRAMEWORK FOR AUTOMATED HOSPITAL MANAGEMENT.

SDG Goals: SDG 3 → Good Health and Well-being (Main Goal)
SDG 9 → Industry, Innovation and Infrastructure
SDG 16 → Peace, Justice and Strong Institutions

Cohort Number: 07

Team Members

USN	NAME
1RR23EC400	Akshay Mallikarjun Andeli
1KN22EC013	Dalia Jahan Liza
4VM22EC011	Dhanushree C H
4VM22EC013	Evangelin
1KN22EC025	Mohammed Auf
1KN22EC027	Muhammad Afnan Sayeed
1RR23EC407	Nagaling Nemati
4MO22EC015	Noor Zaiba
1RR22EC067	Praveen Patil
1VJ22EC009	Rishab H Kadam
4VM22EC047	Sanjana B S
1RR22EC091	Sanju P
1RR22EC109	Sujay S
4MO22EC028	Suzain Khan S
4VM22EC052	Swathi K
1KN22EC055	Usman Khan

Cohort Leaders

NAME	USN	ROLE
Sujay S	1RR22EC109	President
Nagaling Nemati	1RR23EC407	Vice-President
Sanju P	1RR22EC091	Secretary
Akshay Mallikarjun Andeli	1RR23EC400	Deputy Secretary

TABLE OF CONTENTS

LIST OF FIGURES

ABSTRACT

Hospital Management using Radio Frequency Identification (RFID) technology provides a smart and efficient solution to improve identification, access control, and data management in healthcare systems. In this project, RFID cards are assigned to different users such as patients, doctors, staff, and administrators, where each card contains a unique identification number. This unique ID helps in accurately identifying individuals and managing their roles within the hospital environment.

When an RFID card is scanned, the system reads the unique ID and verifies it with the central database. Based on the user type, the system retrieves and displays relevant information such as patient details, medical history, appointment schedules, and doctor information. This process reduces dependency on manual record-keeping, minimizes human errors, and ensures faster access to important data, especially during emergencies.

The system also provides an effective access control mechanism, where only authorized users can access specific data or areas, thereby improving security and privacy within the hospital. Additionally, all activities are recorded and monitored through a dashboard, enabling real-time tracking of user interactions and maintaining proper logs for administrative purposes. This helps hospital management in analyzing data and improving workflow efficiency.

The implementation of this system enhances overall hospital operations by reducing delays, improving accuracy, and ensuring better coordination among patients, doctors, and staff. Although the initial setup may require investment, the long-term advantages such as improved efficiency, better data management, and enhanced patient satisfaction make RFID technology a valuable solution in modern healthcare systems.

Overall, this project demonstrates how RFID-based hospital management systems can create a more secure, reliable, and user-friendly environment, contributing to improved healthcare services and operational effectiveness.

CHAPTER I

INTRODUCTION

1.1 INTRODUCTION

In today's fast-paced healthcare environment, hospitals are expected to provide services that are not only accurate but also quick and efficient. However, many hospitals still rely on traditional methods for managing patients, staff, and records. These manual systems often lead to delays, human errors, and difficulty in accessing important information, which can affect the quality of patient care. As a result, there is a growing need for smart and automated solutions to improve hospital operations.

Radio Frequency Identification (RFID) is a modern technology that allows wireless identification of individuals using unique tags. It has already been successfully used in various fields such as transportation, retail, and logistics. In healthcare, RFID has the potential to simplify processes like patient identification, staff management, and data handling.

This project focuses on developing an RFID-based Hospital Management System that assigns unique RFID cards to patients, doctors, staff, and administrators. When these cards are scanned, the system quickly identifies the user and retrieves relevant information from a central database. This helps in reducing manual work, minimizing errors, and improving the speed of accessing important data.

The system also provides controlled access, ensuring that only authorized users can view or manage specific information. Additionally, a user-friendly dashboard is used to monitor activities and maintain records in real time.

Overall, this project aims to create a more efficient, secure, and reliable hospital management system using RFID technology, ultimately improving hospital workflow and enhancing patient care.

1.2 MOTIVATION

The motivation behind this project stems from real-world challenges faced by hospitals in maintaining accurate records, preventing unauthorized access, and tracking equipment effectively. Many hospitals still rely on manual processes for patient admission, medical file retrieval, and staff attendance. These methods are time-consuming and susceptible to errors, especially during emergencies where timely access to information is critical.

During our observations and research, we noticed that hospitals often deal with issues such as:

- Misidentification of patients
- Loss or misplacement of medical equipment
- Unauthorized entry into restricted areas
- Difficulty in real-time monitoring of staff and assets

1.3 PROBLEM STATEMENT

In many hospitals, the management of patients, staff, and medical records is still carried out using traditional and manual methods. These systems often result in patient misidentification, delays in accessing information, and errors in data handling, which can affect the quality of healthcare services. Additionally, there is a lack of efficient access control, making it difficult to ensure that only authorized personnel can view or modify sensitive information. The absence of a centralized and automated system also leads to inefficient workflow, increased administrative workload, and poor coordination between different departments. In critical situations, the inability to quickly retrieve accurate patient information can lead to delays in treatment and decision-making.

Therefore, there is a need for a smart and reliable system that can automatically identify users, manage data efficiently, and provide secure access control. This project aims to address these challenges by developing an RFID-based Hospital Management System that improves accuracy, reduces manual effort, and enhances overall hospital efficiency.

1.4 OBJECTIVES

- Develop an RFID-based identification system for patients, doctors, staff, and administrators to ensure accurate and fast identification.
- Automate hospital management processes by reducing manual work and minimizing human errors.
- Design a centralized database system for storing and retrieving patient and staff information efficiently.
- Implement secure access control, allowing only authorized users to access specific data or system features.
- Provide a user-friendly dashboard for monitoring, managing, and displaying real-time information.
- Improve overall hospital efficiency by enabling quick access to data and better coordination between departments.
- Enhance patient care and service quality through faster response time and accurate information handling.

1.5 SCOPE OF THE PROJECT

- Develop an RFID-based system for identification of patients, doctors, staff, and administrators
- Enable quick and accurate retrieval of user information using RFID scanning
- Provide a centralized database for storing and managing hospital data
- Include features like:
  - Patient details
  - Appointment management
  - Doctor information
  - User activity monitoring
- Implement basic access control to allow only authorized users to access data
- Design a user-friendly dashboard for real-time monitoring and data display
- Improve hospital workflow efficiency by reducing manual work and errors
- Suitable for small to medium-scale implementation or prototype model

- Limited to basic functionality and does not include:
    - Full hospital system integration
    - Advanced equipment tracking
    - Large-scale deployment
- Can be extended in future with more advanced features and technologies

CHAPTER II

LITERATURE SURVEY

1. **SMART MEDICAL HEALTH CARD FOR HOSPITAL MANAGEMENT**

   This paper presents a Smart Medical Health Card system developed for improving hospital management and patient healthcare services. The proposed system uses a smart card to store patient medical records digitally, making it easier for doctors to access patient history during treatment or emergency situations. The health card contains important details such as patient identification, previous medical reports, prescriptions, and treatment information. The system reduces the need for carrying large physical files and minimizes paperwork inside hospitals. It is especially useful for patients who live alone or are unable to communicate their medical history during emergencies. The implementation improves the speed of medical services and helps healthcare staff provide quick treatment. The system also supports better organization of patient data and reduces the chances of losing medical records. By digitizing healthcare information, the hospital workflow becomes faster and more efficient. The paper highlights the importance of smart healthcare technologies in modern hospitals. However, the proposed system stores only limited information and does not provide real-time patient monitoring or advanced security protection for patient data.

2. **ENHANCING PATIENT HEALTHCARE MANAGEMENT THROUGH RFID TECHNOLOGY**

   This paper explains the use of RFID technology for improving patient healthcare management within hospitals. The system uses RFID tags or wristbands attached to patients for automatic identification and location tracking. RFID readers installed in different hospital areas help doctors and staff monitor patient movement and treatment status in real time. The implementation reduces manual record-keeping and minimizes human errors in patient identification. The system improves hospital workflow management and increases operational efficiency. It also helps healthcare staff quickly access patient information, reducing delays during treatment and emergency situations. RFID technology supports better communication between hospital departments and enhances patient safety. The paper discusses how RFID systems can simplify healthcare operations and improve overall hospital performance. The proposed model also reduces paperwork and helps maintain digital healthcare records effectively. However, the study mentions that successful implementation requires proper collaboration and coordination among all hospital departments, which may be difficult in large healthcare organizations.

3. RFID TECHNOLOGY FOR THE HEALTH CARE SECTOR

This paper focuses on the implementation of RFID technology in healthcare systems for patient monitoring and medical inventory management. The system uses Electronic Patient Codes (EPC) stored in RFID tags to uniquely identify patients and medical equipment. RFID wristbands are assigned to patients, while RFID tags are also attached to medicines and hospital assets for tracking purposes. The implementation helps hospitals maintain accurate patient information and reduces medication errors during treatment. RFID readers automatically collect data from the tags and update hospital records in real time. The system improves healthcare efficiency by enabling faster patient identification and better resource management. It also supports automatic monitoring of drug distribution and patient movement inside hospitals. The paper explains how RFID technology can improve hospital operations and reduce manual work for healthcare staff. The proposed system enhances the accuracy of healthcare services and supports better patient care. However, the research identifies reliability problems and security issues as major challenges in RFID-based healthcare systems, especially regarding patient data privacy and system performance.

4. SMART HEALTH MONITORING AND MANAGEMENT SYSTEM FOR ORGANIZATIONS USING RFID TECHNOLOGY IN HOSPITALS OR EMERGENCY APPLICATIONS

This paper presents a smart health monitoring and management system that combines RFID technology with camera-based monitoring for hospitals and emergency applications. The system records the entry and exit time of patients, visitors, and staff members using RFID cards and card readers. Cameras are used to capture and verify the identity and movement of individuals inside the hospital building. The implementation helps hospitals maintain attendance records and improve security monitoring within the organization. RFID cards are verified using a card screening process before granting access to restricted areas. The system also supports emergency applications by helping hospital authorities monitor activities during critical situations. It improves administrative control and reduces unauthorized access inside healthcare facilities. The proposed model provides better tracking and management compared to traditional manual systems. The paper highlights the usefulness of RFID technology in healthcare monitoring and hospital management applications. However, the system fails to provide accurate real-time tracking of user movement throughout the hospital environment, which limits its effectiveness in large-scale healthcare applications.

5. TRACKING PATIENTS MOVEMENTS USING AN IOT AND RFID SYSTEMS – A CASE STUDY IN A ROMANIA CLINIC

This paper discusses an IoT and RFID-based patient tracking system developed for monitoring patient movement within hospital premises. The system combines RFID technology with IoT communication to collect and transmit patient location data in real time. Each patient is assigned a unique RFID tag that allows hospital staff to monitor their movement inside the clinic. RFID readers installed in different locations capture patient information and send it to the central management system using IoT networks. The implementation improves patient safety and reduces the need for continuous manual supervision by hospital staff. The system also helps doctors and administrators track patient activities efficiently and manage hospital resources effectively. The case study conducted in a Romanian clinic demonstrates how IoT and RFID technologies can improve healthcare monitoring and hospital operations. The paper explains that the proposed model increases healthcare efficiency and supports digital hospital management. It also reduces delays in locating patients during emergencies. However, the system has limitations such as poor fall detection accuracy and lack of advanced 3D tracking support for monitoring patient movement more precisely.

6. HOSPITAL MANAGEMENT SYSTEM USING RFID

This paper proposes a hospital management system developed using RFID technology and mobile application support. RFID tags are assigned to patients for storing identification details and medical information digitally. The system automates patient identification, medication management, and hospital workflow processes. Doctors and hospital staff can use the mobile application to access patient records, monitor treatment details, and manage hospital operations efficiently. The implementation reduces manual paperwork and minimizes human errors in healthcare management. The system also improves communication between hospital departments and supports faster healthcare services. RFID technology helps hospitals track patients and medical resources more accurately. The proposed model increases operational efficiency and improves patient care quality through automation. The paper highlights how digital healthcare systems can simplify hospital administration and reduce workload for staff members. The mobile application allows real-time access to hospital information, improving decision-making and emergency response. However, the system requires stronger encryption techniques and advanced cybersecurity measures to protect sensitive patient information from unauthorized access and security threats.

7. APPLICATION OF RFID SYSTEMS IN HEALTHCARE MANAGEMENT – A SIMULATION FOR EMERGENCY DEPARTMENT

This paper explains the application of RFID systems in healthcare emergency departments for patient monitoring and emergency management. The proposed system uses RFID tags along with motion sensors to monitor patient activities and detect emergency situations such as falls or abnormal movements. When an emergency occurs, the system automatically sends alerts to hospital management and healthcare staff for immediate action. The implementation also provides reminders to patients regarding medication schedules and meal timings. RFID readers continuously collect patient data and support healthcare automation inside emergency departments. The system helps improve response time during critical situations and reduces the burden on hospital staff. The paper highlights the importance of smart healthcare technologies in improving emergency medical services and patient safety. The simulation demonstrates how RFID systems can enhance monitoring efficiency and support better healthcare management. The proposed model also helps hospitals maintain patient records digitally and improve healthcare coordination. However, the study mentions that the system suffers from lower accuracy in emergency detection and monitoring, which affects the reliability and overall performance of the healthcare system.

8. HOSPITAL MANAGEMENT SYSTEM USING RFID TECHNOLOGY

This paper presents an RFID-based hospital management system integrated with a dedicated mobile application for hospital staff and administrators. RFID tags are used for patient identification, tracking, and digital healthcare record management. The mobile application allows administrators and healthcare staff to monitor hospital activities remotely and receive real-time alerts regarding patient conditions and emergencies. The implementation improves communication between hospital departments and enhances operational efficiency. The system reduces paperwork and supports digital management of hospital resources and patient information. It also enables quick access to patient records, helping doctors provide faster treatment and better healthcare services. The paper discusses how RFID technology combined with mobile applications can improve healthcare automation and simplify hospital management tasks. The system supports real-time monitoring and remote access to hospital data, making hospital operations more efficient and organized. The implementation also improves patient safety and healthcare service quality. However, the proposed system faces challenges related to scalability and data security, especially when handling large amounts of patient information in large hospital environments.

9. RFID TAGS BASED HOSPITAL MANAGEMENT SYSTEM

This paper introduces a hospital management system based on RFID tags for organizing and managing patient information efficiently. Each patient is assigned a unique RFID tag containing medical records and identification details. RFID readers are used to collect patient information automatically and update hospital records digitally. The system improves patient identification accuracy and reduces manual paperwork in hospitals. It also helps healthcare staff access patient information quickly during treatment and emergency situations. The implementation supports efficient patient monitoring and simplifies hospital administration tasks. The paper explains that RFID technology improves healthcare management by reducing errors and enhancing workflow efficiency. The proposed system also helps hospitals maintain digital patient databases for better healthcare services. The implementation minimizes delays in accessing patient information and improves coordination between hospital departments. However, the study identifies limitations such as lack of immediate data retrieval and insufficient information storage capacity, which reduce the overall efficiency of the system.

10. HOSPITAL MANAGEMENT SYSTEM USING RFID

This paper describes a hospital management system developed using RFID technology for digital patient record management. The system assigns unique RFID tags to patients for storing healthcare information electronically. RFID readers are used to retrieve patient data quickly and support faster treatment processes within hospitals. The implementation reduces the need for physical medical files and improves healthcare workflow management. The system also supports digital prescription generation, helping doctors and healthcare staff manage patient treatment efficiently. RFID technology enables automatic patient identification and minimizes manual errors in healthcare operations. The proposed model improves patient care quality and increases the efficiency of hospital administration. The paper highlights the importance of digital healthcare systems in modern hospitals for reducing paperwork and enhancing healthcare automation. The implementation supports faster communication between departments and better organization of patient records. However, the study mentions that the system lacks strong security mechanisms to protect sensitive patient information, which may lead to unauthorized access and data privacy issues.

CHAPTER III
METHODOLOGY

3.1 METHODOLOGY

The development of the RFID-based Hospital Management System was carried out using a structured and modular methodology, ensuring each subsystem functioned independently before being integrated. This method facilitated easier debugging, testing, and refinement throughout the project.

1. **System Design Planning**

   The first step involved identifying the key functions required in the hospital environment such as patient identification, access control for medical staff, equipment access tracking, and prescription data retrieval.

   Based on these, appropriate components were selected:

   - ESP32 was chosen for its dual functionality of microcontroller and Wi-Fi connectivity.
   - RC522 RFID Reader and RFID passive tags (cards/key fobs) were selected for identity verification.
   - Servo motor for gate/door simulation.
   - LEDs for access indication.
   - A central web dashboard was proposed to monitor and display RFID access events and patient data.

2. **Hardware Integration**

   The hardware components were connected and tested in stages:

   - The RC522 RFID Reader was connected to the ESP32 via SPI protocol.
   - RFID cards and key fobs were assigned unique patient/staff IDs.
   - The servo motor was connected to simulate gate opening when access was granted.
   - LEDs were used to indicate access status (green for allowed, red for denied).
   - A regulated power supply ensured stable operation of all components.
   - PIR Sensor were connected for motion sensor.

- LCD were connected for displaying the process.

Each component was tested individually and then integrated into a single working prototype, mounted on a board for easy demonstration.

3. Embedded Programming

Using Arduino IDE, the ESP32 was programmed in Embedded C.

The logic included:

- Reading RFID UID from the reader.
- Verifying UID against predefined entries in memory.
- Controlling the servo motor to open the gate for authorized users.
- Triggering LEDs and serial print statements for feedback.
- Sending UID data via serial communication to the computer.

This code formed the heart of the embedded system controlling access operations in real-time.

4. System Testing and Validation

After development, the system underwent several testing phases:

- **Functional Testing:** Each RFID tag/card was tested to ensure proper UID reading and matching.
- **Access Testing:** Authorized tags opened the gate and displayed details; unauthorized tags were blocked.
- **Communication Testing:** Verified real-time serial data exchange between ESP32 and the dashboard.
- **Power Testing:** The system was evaluated under varied input voltages to test power regulation.

Each phase helped refine the logic and user interface to ensure accuracy, stability, and responsiveness.

5. Deployment Setup

The finalized setup included all components mounted neatly for demonstration. Wires were routed properly, and a clean interface was shown on the screen. Photographs of this implementation have been embedded in the previous section of this chapter for visual understanding.

3.2 ALGORITHM

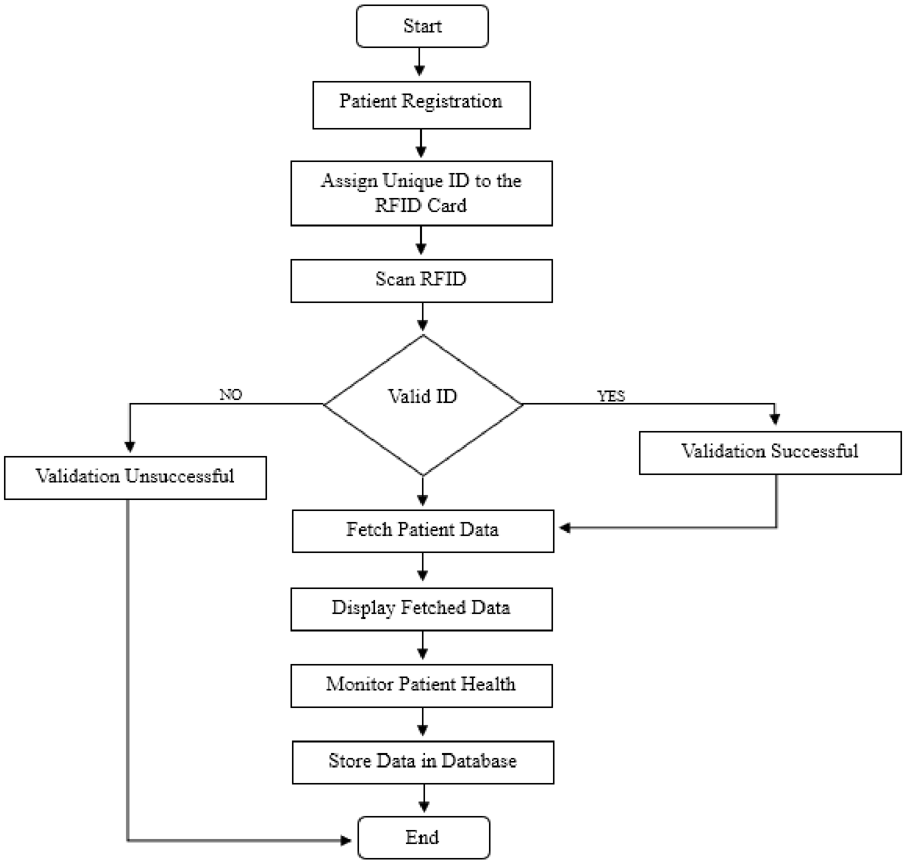

Figure a: Flowchart

Step 1 → Start the system and initialize ESP32, RFID reader, and web interface.

Step 2 → Scan the RFID card/tag when a patient, staff, or equipment is detected.

Step 3 → Read the Unique RFID ID from the card/tag.

Step 4 → Check the scanned ID in the hospital database.

Step 5 → • If patient ID is found → Display patient details on the web and log the entry.

• If staff ID is found → Verify access permissions and control the entry-exit gate using a stepper motor.

• If equipment ID is found → Update the tracking status in the system.

• If ID not found → Deny access and trigger a buzzer alert.

Step 6 → Send the scanned data to the web-based system for real-time monitoring.

Step 7 → Update the database with the latest entry logs, staff movements, and equipment tracking.

Step 8 → Repeat steps 2 to 7 for continuous monitoring.

Step 9 → Stop the system when manually turned off.

CHAPTER IV

HARDWARE AND SOFTWARE REQUIREMENTS

4.1 HARDWARE DESCRIPTION

1. ESP32

Figure b: ESP32 Microcontroller

The ESP32 is a highly versatile and powerful 32-bit microcontroller developed by Espressif Systems. It's one of the most popular choices for developing IoT applications due to its built-in Wi-Fi and Bluetooth capabilities, which allow devices to communicate wirelessly. The ESP32 features a dual-core processor, which is capable of handling multiple tasks simultaneously, making it suitable for applications that require real-time processing and multitasking.

The microcontroller is also known for its flexibility, as it supports a wide range of communication protocols like I2C, SPI, UART, and PWM. These features allow it to interface with various peripherals and sensors seamlessly, making it a perfect choice for complex applications like smart systems and IoT devices. ESP32 modules come with various integrated peripherals, including digital-to-analog converters (DAC), analog-to-digital converters (ADC), and Pulse Width Modulation (PWM) capabilities, giving developers flexibility in interfacing with sensors, motors, and displays.

The ESP32 is well-suited for use in low-power IoT applications, thanks to its power management features, such as deep sleep mode (which consumes as low as 10 µA of current). This makes it ideal for battery-operated projects or systems where energy efficiency is a priority. With integrated hardware encryption and secure boot options, the ESP32 offers a high level of security, which is critical for IoT devices, especially in healthcare applications where privacy and data integrity are vital.

2. RFID Reader (RC522)

Figure c: RFID Reader (RC522)

The RC522 is a compact, low-cost 13.56 MHz RFID module designed for reading RFID tags that adhere to the ISO/IEC 14443A standard. It is based on the MFRC522 chip, which supports a wide variety of RFID tags and smart cards. The module works by generating an electromagnetic field that powers the RFID tags and allows them to communicate with the reader. This contactless communication makes the RC522 module ideal for applications such as access control systems, inventory tracking, and asset management.

The RC522 module operates on the 13.56 MHz frequency, which is a global standard for RFID technology. This frequency is used for various applications such as personal identification, contactless payments, and ticketing systems. The RC522 can communicate with tags in close proximity (typically within 3-5 cm), which makes it useful in environments where security and quick access are required. The module is compact and easy to integrate with microcontrollers like the ESP32 via the SPI interface, which offers fast communication speeds for real-time operations. One of the key advantages of the RC522 is its low power consumption. It can be powered by a 3.3V or 5V power supply, and its low current draw makes it suitable for battery-operated systems. Additionally, the module is highly affordable, which makes it a popular choice for DIY projects and commercial applications. Its read-write capabilities allow it to interact with various types of RFID tags, from simple MIFARE cards to more complex RFID tags used in asset tracking or healthcare management applications.

3. RFID Passive Inlays

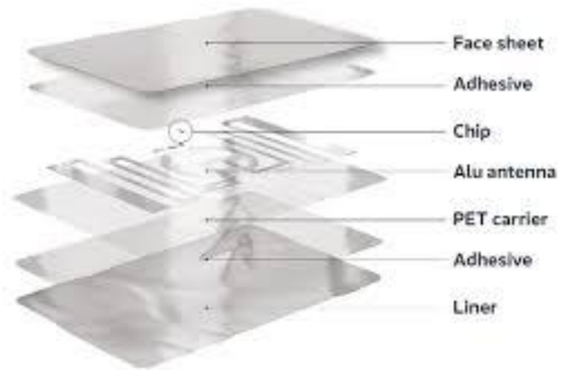

Figure d: RFID Passive Inlay

RFID passive inlays are ultra-thin components made of an integrated circuit (IC) attached to a copper or aluminium antenna, all mounted on a flexible substrate (usually PET). These inlays are typically embedded within labels or inserted into plastic products to convert them into smart tags. Being "passive," they do not contain a battery but are powered by the electromagnetic field of the RFID reader. Passive inlays are the core of any RFID tag product they are highly customizable and used in applications ranging from supply chain management and retail to healthcare and pharmaceuticals.

4. RFID Card

Figure e: RFID Card

RFID cards are smart cards with an embedded passive RFID inlay inside a PVC shell, commonly used for access control, identity verification, and transaction processing. They are the size of a credit card and can be easily carried in a wallet. These cards operate using radio waves and can be scanned quickly, even without direct line-ofsight, making them efficient and secure in high-traffic environments like hospitals, offices, and campuses. Most RFID cards are rewritable and can store user-specific data.

5. RFID Key Fob

Figure f: RFID Key fob

RFID key fobs are small, durable tokens that contain the same electronics as RFID cards but in a rugged plastic case designed to hang on keyrings. These are particularly useful in applications requiring frequent handling, portability, and durability. Their ergonomic design and sturdy build make them ideal for long-term use by patients and visitors. The fob contains an embedded coil and chip, just like cards, but often in a rounded or teardrop shape.

6. Servo Motor (SG90)

Figure g: Servo Motor (SG90)

The SG90 is a small and lightweight servo motor designed for precise control over angular movement. It has a range of motion between 0° and 180° and can be controlled via a PWM signal, allowing it to rotate to specific angles. Servo motors are commonly used in applications requiring high accuracy and quick response, such as robotics, automated systems, and access control systems.

The SG90 is a high-precision servo motor that is often used in small-scale projects and IoT devices due to its compact size and lightweight design. It is powered by a 5V supply and can be controlled using PWM signals from microcontrollers like the ESP32.

7. **Power Supply**

Power supply is necessary for providing stable voltage and current to the system components. In embedded systems like the Hospital Management System, a 5V DC power supply is used to power components like the ESP32, RC522 RFID reader, servo motor, and LEDs. The power supply ensures that each component operates correctly and efficiently, preventing voltage fluctuations that could lead to instability or failure.

4.2 SOFTWARE DESCRIPTION

1. **Arduino IDE**

The Arduino IDE (Integrated Development Environment) is an open-source application designed for writing, compiling, and uploading embedded C/C++ code to Arduino and compatible microcontrollers. It simplifies hardware-software interaction and is widely used in academic, hobbyist, and prototyping environments. Arduino IDE supports embedded C/C++ programming and includes a simple code editor with features such as syntax highlighting, automatic indentation, and error notifications. It allows users to manage libraries, interface with hardware through built-in functions, and monitor serial communication in real-time using the Serial Monitor. The IDE provides support for third-party board definitions, enabling it to work with a range of microcontrollers including the ESP32. The integrated bootloader simplifies firmware uploading over USB, while the vast online community and library support allow for quick integration of modules like RFID, Servo, and SPI.

In our project, the Arduino IDE is used to develop and upload the control program to the ESP32 board. The embedded code handles RFID tag reading using the MFRC522 module, servo motor control for gate access, LED status indication, and serial communication with the PC. The Serial Monitor is utilized during development for real-time debugging of tag reads and logic flow.

2. Visual Studio Code

Visual Studio Code (VS Code) is a lightweight, yet powerful source code editor developed by Microsoft. It supports a wide array of programming languages and frameworks and is highly extensible through its marketplace plugins. It is especially popular among developers for Python, web development, and embedded systems integration. VS Code offers a highly customizable development environment with features such as IntelliSense (smart code completion), syntax highlighting, Git integration, terminal access, and debugging tools. It supports extensions for Python, HTML, CSS, and serial port monitoring, making it ideal for both backend and frontend development. Real-time collaboration is also possible via Live Share. With Python extensions, VS Code enables execution of scripts for serial communication, data logging, and GUI updates. The editor's file management and workspace feature also make it convenient for handling multi-file projects. In our hospital management system, VS Code is used as the central development platform for both backend Python code and the frontend HTML/CSS interface. Python scripts written here read serial data from the ESP32, identify RFID tags, and update hospital access logs. The HTML and CSS code, also written in VS Code, forms the user dashboard interface displaying login information and patient data dynamically based on tag reads.

3. HTML (HyperText Markup Language) and FLASK

HTML (HyperText Markup Language) is the standard language used to design and structure web pages. It is used to create the layout of a website by defining elements such as headings, paragraphs, tables, forms, and buttons. In this project, HTML is used to develop the user interface of the hospital management system, where patient details, appointment information, doctor data, and other records are displayed in a clear and organized manner. It provides a simple and user-friendly way for users to interact with the system.

Flask is a lightweight web framework in Python that is used to build web applications and handle the backend operations. It acts as a bridge between the frontend (HTML) and the database. In this project, Flask is used to process the data received from the system, retrieve information from the database, and send it to the HTML pages for display. It also handles routing, user requests, and data processing.

CHAPTER V

SYSTEM DESIGN

5.1 BLOCK DIAGRAM

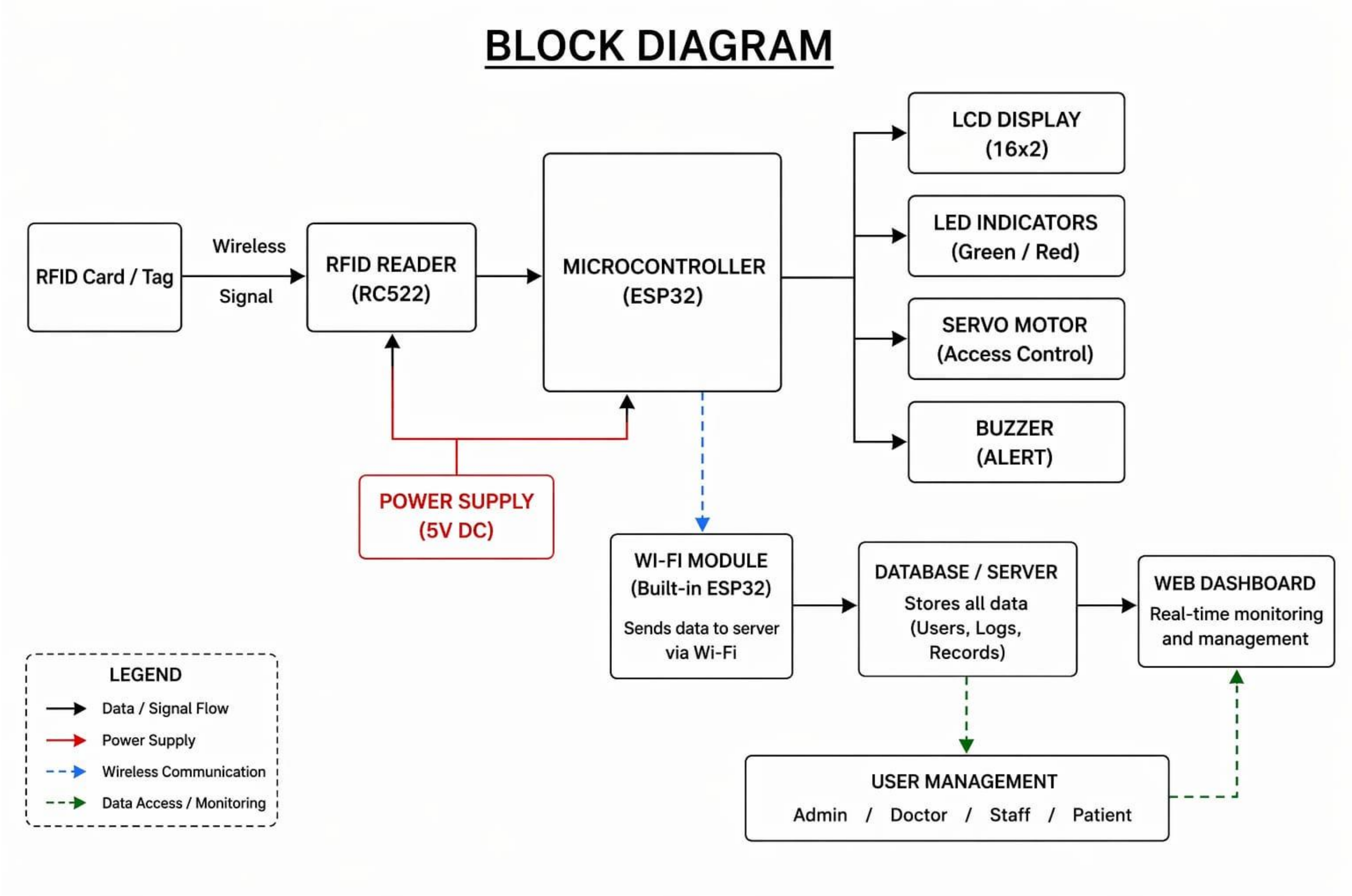

Figure h: Block Diagram

5.2 PROPOSED SOLUTIONS

The Hospital Management System using RFID is designed to overcome inefficiencies in traditional hospital operations by integrating RFID technology, an ESP32 microcontroller, and a web-based monitoring platform. Each patient is assigned an RFID card, enabling automated identification and instant access to medical records, reducing manual paperwork and errors. Similarly, hospital staff will use RFID-enabled ID cards for secure authentication and controlled access to restricted areas, managed through a stepper motor-controlled entry-exit system. To prevent misplacement and optimize resource utilization, RFID tags will be attached to medical equipment, allowing for real-time tracking and theft prevention. Additionally, the system will automate medicine and inventory management, ensure timely restocking and minimize supply shortages. The integration of ESP32 facilitates wireless communication with a web-based hospital management system, providing real-time data on patient records, staff movements, and medical equipment tracking.

By implementing this system, hospitals can enhance security, improve operational efficiency, reduce human errors, and ensure smooth workflow management. The solution offers a cost-effective and scalable approach to modernizing hospital operations through RFID-based automation and digital monitoring.

IMPLEMENTATION

A prototype model was developed to validate the Hospital Management System using RFID technology. The setup focuses on patient data management, staff access control, secure pharmacist access to prescriptions, and equipment data management. The central component of the system is the ESP32 microcontroller, which interfaces with three RC522 RFID readers via the SPI communication protocol. These readers are responsible for scanning RFID tags on patients, staff, and equipment. The servo motor is used for staff access control, unlocking doors when authorized RFID tags are detected. The RFID system also handles patient data management by tracking patient identification and medical history stored on RFID tags. For secure pharmacist access to prescriptions, only authorized RFID tags allow access to prescription data, reducing the risk of medication errors. Equipment data management is achieved by tagging hospital assets with RFID, enabling real-time tracking of equipment location and usage. The system is connected to a laptop hosting the web application interface, which allows real-time monitoring of patient data, staff access, prescription logs, and equipment status. The application provides an interactive platform for managing hospital operations efficiently and securely.

CHAPTER VI

RESULTS

The system was rigorously tested under various scenarios to evaluate its performance. The RFID readers successfully identified authorized tags, leading to servo motor actuation and granting access. Unauthorized tags were denied access, ensuring system reliability and security.

The developed web dashboard efficiently captured and displayed real-time data including staff login credentials, patient information, and entry-exit logs.

Key observations include:

- Successful RFID authentication within 1–2 seconds of scanning.
- Real-time updating of staff attendance and patient data.
- Reliable servo motor actuation for entry control without noticeable delay.
- Pharmacists were provided with RFID-based access to patient prescriptions through the web dashboard.
- Equipment data was managed through RFID tags, indicating whether a device was operational or under maintenance.

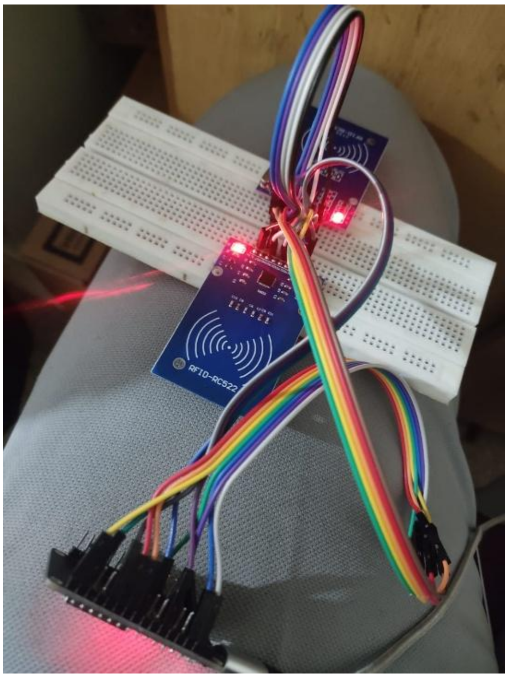

Figure i: Model

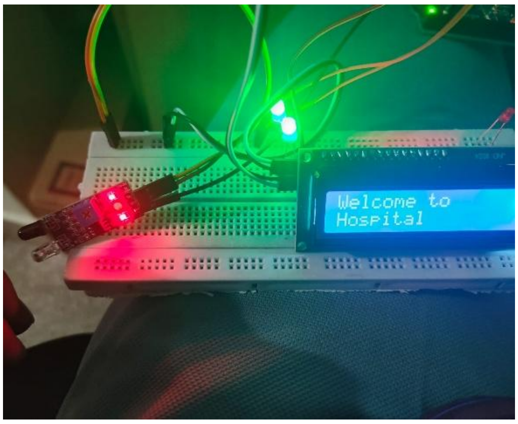

Figure j: IR Sensor with LCD

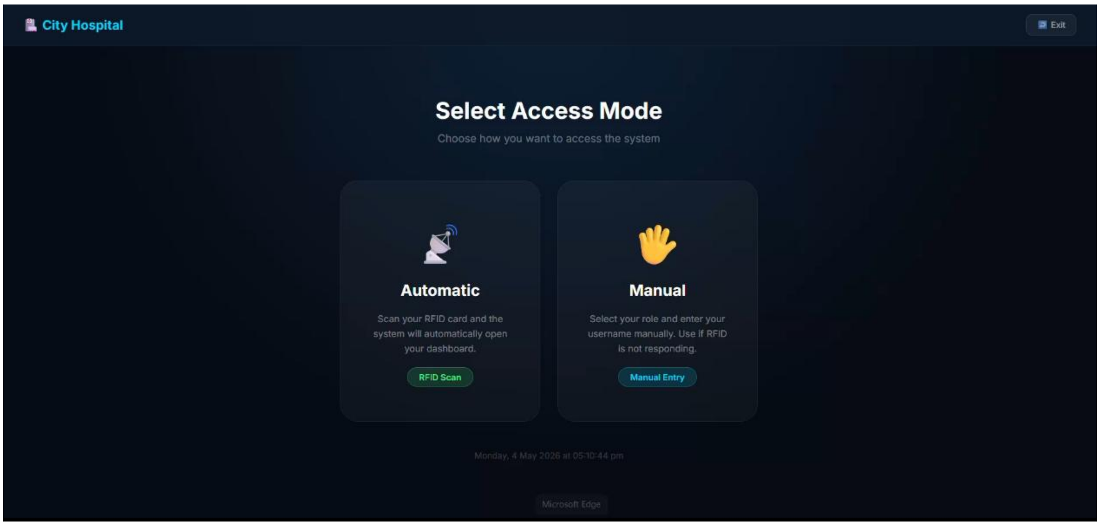

Figure k: Webpage

The above interface represents the Access Mode Selection page of the RFID-Based Hospital Management System. This page is designed to provide users with two different methods for accessing the hospital dashboard system: Automatic Mode and Manual Mode. The interface uses a dark-themed modern dashboard layout to improve visibility and provide a professional healthcare environment appearance.

In the Automatic Mode section, users can access the system by scanning their RFID card. Once the RFID card is scanned, the system automatically identifies the user and redirects them to the corresponding dashboard based on their role, such as doctor, nurse, staff, or administrator. This feature helps reduce manual work, improves authentication speed, and enhances security within the hospital environment.

The Manual Mode section allows users to log in manually by selecting their role and entering their username or credentials. This option acts as a backup method when the RFID scanner is unavailable or not responding. By providing both automatic and manual access methods, the system ensures reliability, flexibility, and uninterrupted operation in hospital management processes.

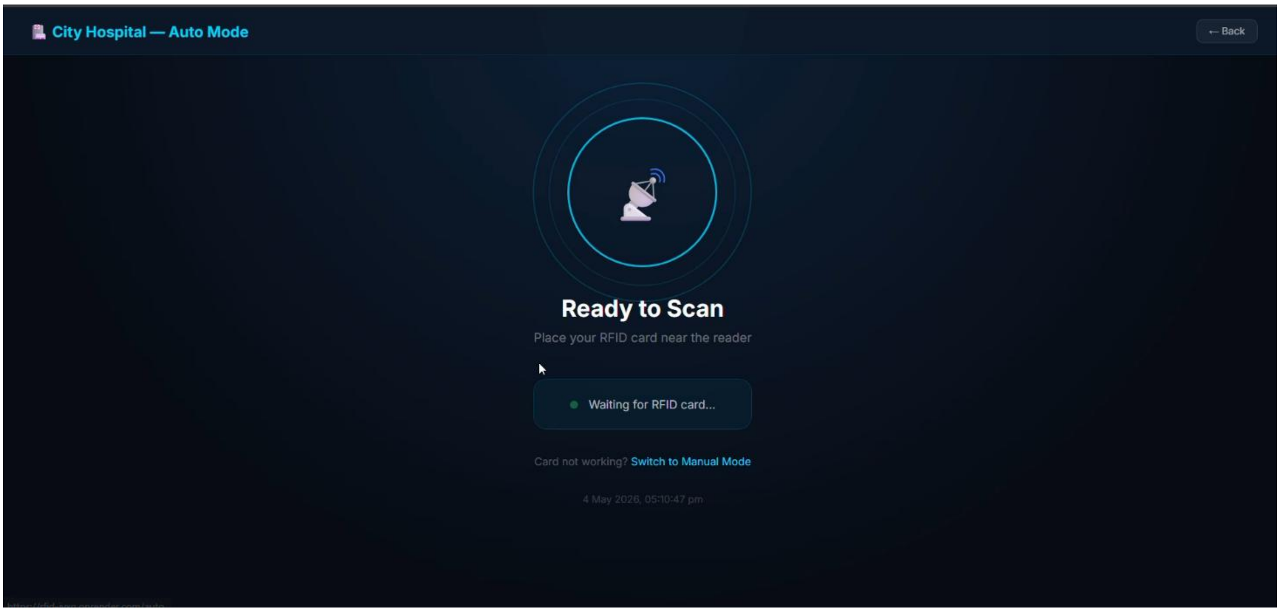

Figure l: RFID Scanning Page

The above image represents the Automatic RFID Scanning Interface of the RFID-Based Hospital Management System. This interface is designed to enable fast and secure user authentication through RFID technology. The page displays a “Ready to Scan” message, indicating that the system is actively waiting for an RFID card to be scanned near the RFID reader.

When a valid RFID card is detected, the system automatically identifies the registered user and redirects them to their respective dashboard based on their assigned role, such as doctor, nurse, staff, or administrator. This automated authentication process minimizes manual login procedures, reduces time consumption, and improves operational efficiency within the hospital environment.

The interface also includes a real-time status indicator showing “Waiting for RFID card…”, which confirms that the RFID reader is active and functioning properly. Additionally, a “Switch to Manual Mode” option is provided as an alternative login method in case the RFID card is not working or the reader encounters issues. Overall, this module enhances convenience, security, and automation in hospital access management.

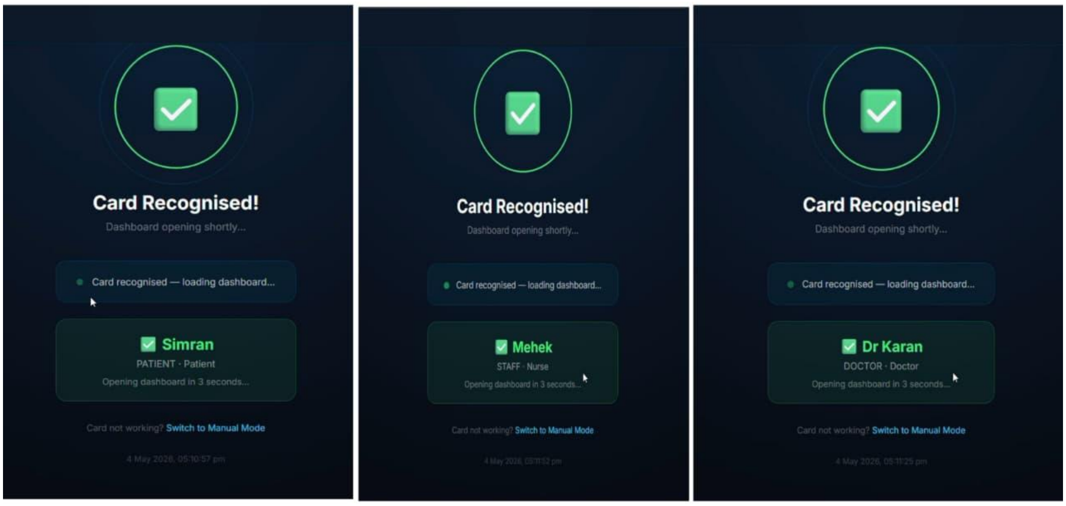

Figure m: RFID Recognized Page

The above image demonstrates the RFID Card Recognition Process in the RFID-Based Hospital Management System. This interface appears after an RFID card is successfully scanned and verified by the system. The page confirms authentication by displaying the message "Card Recognised!" along with the user's name and role information.

The first interface shows a patient account named Simran, the second interface displays a staff/nurse account named Mehek, and the third interface represents a doctor account named Dr. Karan. Based on the RFID card data stored in the centralized database, the system automatically identifies the user category and redirects them to the appropriate dashboard. This role-based access mechanism ensures that each user can only access functionalities relevant to their responsibilities within the hospital.

The interface also displays a loading status indicating that the dashboard will open automatically within a few seconds. This automated process improves efficiency, minimizes manual login efforts, and enhances security by preventing unauthorized access. Overall, the RFID recognition module enables quick, reliable, and user-friendly authentication for different hospital personnel and patients.

CHAPTER VII
CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The RFID-based Hospital Management System developed in this project provides an effective solution to improve patient identification, access control, and data management in hospitals. By assigning unique RFID IDs to patients, doctors, staff, and administrators, the system ensures accurate and fast identification, reducing the chances of human errors associated with manual processes.

The integration of a centralized database and user-friendly dashboard enables quick retrieval and monitoring of important information such as patient details, appointments, and doctor data. This improves workflow efficiency and helps healthcare professionals make faster and better decisions.

The system also enhances security by allowing only authorized users to access specific data, ensuring proper data handling and privacy. Although the project is implemented at a prototype level, it clearly demonstrates the potential of RFID technology in modernizing hospital operations.

Overall, the project successfully shows how the use of RFID technology can create a more efficient, reliable, and user-friendly hospital management system, ultimately contributing to improved healthcare services and patient satisfaction.

7.2 FUTURE SCOPE

The proposed Hospital Management System using RFID Technology offers several opportunities for future enhancement and scalability:

- **Cloud Integration:** Future versions can integrate cloud-based storage to allow secure, centralized access to patient records and logs across multiple hospital branches.

- **Mobile Application Support:** A dedicated mobile app can be developed for staff and administrators to view real-time data, receive alerts, and manage hospital operations remotely.
- **Biometric Authentication:** Combining RFID with biometric verification (e.g., fingerprint or facial recognition) can further strengthen access control and identity validation.
- **AI-Based Predictive Maintenance:** Using AI algorithms, equipment data can be analyzed to predict maintenance needs before failures occur, improving hospital efficiency and safety.
- **Integration with National Health Portals:** Linking with government or insurance health databases could streamline patient record management and facilitate automated billing.

CHAPTER VIII

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